

VISTAJOBS PLATFORM

Software Development Document - Development to QA Handover

Project: VistaJobs - recruitment, verification, employer matching and administration platform

Prepared for: QA Testing and TL Review

Prepared by: Development Team

Version: 1.0

Date: July 2026

Status: Ready for QA validation after backend/frontend deployment

1. Purpose

This document gives QA engineers a concise technical and functional overview of the VistaJobs platform so they can perform functional, integration, regression, security, responsive UI and UAT testing without reading the full codebase first.

2. Application Overview

VistaJobs is an ASP.NET Core Web API plus static frontend recruitment platform. It supports jobseeker registration, role-based login, fresher and experienced candidate profiles, Aadhaar/PAN/UAN verification, resume upload, employer registration with company validation, employer candidate matching, job postings, applications and admin dashboards.

3. Technology Stack

Layer	Technology / Implementation
Backend	ASP.NET Core Web API (.NET 8), C#
Frontend	HTML, CSS, JavaScript static frontend under Frontend/
Database	SQL Server with Entity Framework Core migrations
Authentication	JWT bearer tokens, email OTP, BCrypt password hashing, optional Google sign-in
Verification	Local format validation plus DigiLocker/Surepass EPFO integration points
Deployment	Azure App Service API, Azure Static Apps or local Live Server frontend

4. Architecture

Browser frontend -> API controllers -> services/providers -> Entity Framework Core -> SQL Server. Resume files are uploaded to the API and served through the /Uploads path. The frontend stores the authenticated session in sessionStorage and routes users by the role returned from login.

Component	Primary Files	QA Focus
API startup	Program.cs, appsettings*.json	CORS, JWT configuration, uploads, Swagger and seed data

Component	Primary Files	QA Focus
Controllers	Auth, Candidates, Jobs, Applications, Admin, Verification	Validation, HTTP status codes, authorization and response shape
Models/Data	Models/*, ApplicationDbContext.cs, Migrations/*	Schema mapping, duplicate prevention and update behavior
Frontend	Frontend/index.html, js/app.js, css/*	Role routing, forms, toasts, responsive states and API calls

5. User Roles

Role	Expected Access	Must Not Access
Jobseeker	Register, login, create/update fresher or experienced profile, upload resume, verify Aadhaar/PAN/UAN, apply for jobs	Employer matching dashboard, admin lists
Employer	Register through employer form, login, enter matching requirements, view matching candidates, create jobs	Sensitive verification APIs outside allowed matching view, admin dashboards
Admin	Dashboard summary, raw users, jobs and applications lists	No unauthorized access without admin role

6. Core Modules

Module	Responsibility	Key Test Areas
Authentication	Register/login via email OTP, Google auth guard, JWT issuance	Required fields, invalid credentials, OTP expiry, role routing
Jobseeker Profiles	Fresher/experienced candidate details and resume upload	Create, update, validation, duplicate profile by email
Verification	Aadhaar, PAN, UAN and verification status	Format validation, provider fallback, status persistence
Employer Registration	Company name, official email, GST, CIN, website and OTP	Personal email blocking, domain/website match, OTP verification
Employer Matching	Requirement entry and skill-based candidate matching	Skill aliases, candidate type, experience sorting, location preference
Jobs and Applications	Job posting and candidate application tracking	CRUD, duplicate application blocking, confirmation email
Administration	User/job/application visibility for admins	Role authorization and dashboard counts

7. End-to-End QA Workflows

7.1 Jobseeker Workflow

1. Register as a jobseeker using name, email, password and email OTP.
2. Login with the same email and password, complete login OTP, and confirm jobseeker dashboard routing.
3. Submit or update fresher/experienced profile and confirm `/api/Candidates/my-profile` returns saved data.

4. Upload PDF/DOC/DOCX resume and confirm the generated /Uploads path opens.
5. Run Aadhaar, PAN and UAN verification flows and verify status indicators.
6. Apply for a job and confirm duplicate application is blocked.

7.2 Employer Workflow

1. Register through employer form and verify personal email domains are blocked.
2. Complete company identity fields and OTP, then login as employer.
3. Enter role, candidate type, skills, experience and location to find matching candidates.
4. Create a job posting and verify admin/job APIs reflect it.

7.3 Admin Workflow

1. Start the API and confirm the default admin account is created or updated.
2. Login with the seeded admin account provided by the development team.
3. Verify dashboard counts and raw lists for users, jobs and applications.
4. Confirm non-admin roles cannot open admin APIs or admin frontend views.

8. Critical Business Rules

- Protected APIs require JWT authentication and role checks where applicable.
- Registration and login flows require OTP verification before account/session completion.
- Employer registration must reject personal email domains and validate company identity fields.
- Candidate profile is keyed by login email and should update, not duplicate, for the same user.
- Employer matching must use only relevant candidate profile fields and must not expose unrelated sensitive data.
- Duplicate job applications must be blocked by job ID and candidate email.
- Resume upload accepts only PDF, DOC and DOCX files.

9. QA Scope

Area	In Scope Checks
Functional	Registration, login, OTP, role routing, profiles, verification, matching, jobs, applications and admin dashboards
Validation	Required fields, email format, official email policy, GST/CIN length, website matching, password rules and upload types
Integration	SQL Server persistence, SMTP OTP/email, verification provider configuration, static frontend/API CORS
Security	JWT requirement, role isolation, no sensitive data in public pages, session clearing on logout
Responsive UI	Chrome, Edge and Firefox on desktop plus mobile width smoke tests
Regression	Retest login, profile save, employer matching and admin after every deployment

10. Environment Summary

Environment Item	Value / QA Note
Local frontend	http://127.0.0.1:5500/index.html
Local API	https://localhost:7250/api or http://localhost:5295/api based on launch profile
Swagger	https://localhost:7250/swagger
Database	SQL Server via ConnectionStrings:DefaultConnection
Secrets	Use user-secrets locally and Azure App Service configuration in hosting. Do not share secrets in QA artifacts.
Migration check	GET /__migrations

11. Deployment and Configuration Checks

- Confirm JWT issuer, audience and key are configured before starting the API.
- Confirm CORS allows local frontend and Azure Static Apps origin.
- Confirm frontend API base URL points to local API for local testing and Azure API for deployed testing.
- Confirm Uploads/ is writable and uploaded resume links are reachable.
- Confirm SMTP credentials are valid before OTP, reset password and application email testing.
- Confirm Google OAuth authorized origins before enabling Google sign-in in deployed environments.

12. High-Level Testing Checklist

Done	Checklist Item	Notes
☐	Role-based access and logout	
☐	Jobseeker registration and login OTP	
☐	Employer registration and official email validation	
☐	Admin seed and admin dashboard	
☐	Fresher and experienced profile save/update	
☐	Resume upload and public link access	
☐	Aadhaar, PAN and UAN verification	
☐	Employer matching candidate results	
☐	Job creation and application duplicate prevention	
☐	Validation messages and toast behavior	
☐	CORS and API error handling	
☐	Responsive UI and browser compatibility	
☐	Regression testing before release	

13. Risks and Assumptions

- Email OTP and notification testing depends on valid SMTP settings and mailbox access.
- Verification provider behavior depends on DigiLocker/Surepass settings; local fallback behavior must be documented for QA.
- Production-like testing should use isolated QA data and must not use real Aadhaar/PAN/UAN values unless approved.
- Azure deployment must update both API and frontend; deploying only the frontend can leave old CORS/API behavior active.
- Google sign-in requires the deployed origin to be added in Google Cloud OAuth settings.

14. Version History and Approval

Version	Date	Description
1.0	July 2026	Initial VistaJobs Development-to-QA handover
1.1	Future	Update after QA feedback, deployment changes or new module additions

Prepared By: Development Team

Reviewed By: _____

Approved By: _____

QA Sign-off Date: _____